

馬韋寧 (WeiWei Ma)

Email: vernessa0607@gmail.com

Location: Taiwan (GMT+8)

GitHub: github.com/weiwei0607

SUMMARY

Junior Full-Stack Developer with a **biomedical engineering background** and strong product-building mindset. Independently designed, architected, and shipped **20+ applications** spanning health tech, AI-integrated tools, productivity apps, and interactive entertainment. Proficient in rapid prototyping using Flutter, React, and Python — **capable of going from concept to deployed MVP within days**. Passionate about building technology that solves real problems, with a **unique edge at the intersection of healthcare domain knowledge and modern software engineering**.

TECHNICAL SKILLS

Mobile Development - Flutter (Dart): Cross-platform iOS/Android apps, state management, Firebase integration

Frontend Development - React, TypeScript, Vite, HTML5/CSS3 - Responsive design, PWA (Progressive Web App) implementation - UI/UX prototyping and component architecture

Backend & API Development - Python: FastAPI, Flask, automation scripting - Node.js: Express, RESTful API design - Firebase: Authentication, Firestore, Cloud Functions, Hosting

Database & Data Management - SQLite, PostgreSQL - Data modeling for small-to-medium scale applications

AI Integration & LLM Applications - OpenAI API (GPT-4, GPT-4o), Google Gemini API - LLM application architecture: prompt engineering, context management - Building AI-powered features: chatbots, content generation, sentiment analysis

DevOps & Tools - Git version control, GitHub Actions (basic CI/CD) - Docker (containerization basics) - Google Cloud Platform (Cloud SDK)

Product & Planning - Product requirement documentation, user flow design - Technical architecture planning (API design, database schema) - Agile development approach

FEATURED PROJECTS

✂️ B-Battle — Gamified Budgeting RPG

Tech Stack: React 19, Vite, Firebase, Zustand, Google Gemini API, Tailwind CSS v4

Live Demo: willpower-battle.web.app

GitHub: github.com/weiwei0607/b-battle

- Reimagined budgeting as an RPG: spending depletes an HP bar, over-budget triggers "battle loss" state
- Built an AI companion system with 3 selectable personalities (Peer / Mentor / Nemesis) powered by Gemini API, delivering real-time contextual commentary on each transaction
- Designed achievement and equipment systems to reward consistent budget adherence
- Firebase Firestore handles real-time sync across devices with Auth for multi-user support
- Demonstrates: React 19 state architecture (Zustand), Gemini API integration, gamification UX design, Firebase backend

🗺️ History Bridge — Historical Figure Connection Game

Tech Stack: React 18, Vite, TypeScript, BFS Algorithm

GitHub: github.com/weiwei0607/history-bridge

- Built a "Six Degrees of Separation" game for 500+ historical figures spanning ancient Greece to modern era
- Implemented BFS (Breadth-First Search) pathfinding algorithm to compute shortest relationship chains between any two figures
- Designed two game modes: path exploration (find the connection) and timed quiz (identify figures from clues)
- Relationship database covers 6 connection types: teacher-student, rivals, contemporaries, influence, family, collaboration
- Demonstrates: graph algorithm implementation, TypeScript, educational game UX design

✂️ Task Shredder — AI Task Breakdown Tool

Tech Stack: Next.js, Google Gemini API, Notion API, Google OAuth, Framer Motion

GitHub: github.com/weiwei0607/task-shredder

- "Brain dump" to structured tasks: users input unorganized thoughts and Gemini API categorizes, clarifies, and breaks them into actionable sub-tasks
- Designed clarifying dialogue flow: AI asks targeted questions before generating breakdown, reducing vague output
- Notion as persistent backend for task storage; Google OAuth for authentication; ics export for Google Calendar sync
- Mermaid diagram visualization for task dependency mapping
- Demonstrates: full-stack Next.js (App Router), multi-API integration, conversational AI design

ADDITIONAL PROJECTS

Project	Tech	Description
Stock Analyzer	Python, FastAPI, APScheduler	Personal stock tracker with Telegram Bot alerts, MA/RSI/MACD technical analysis
Negativity Absorber	Flutter, OpenAI API, Firebase	Cross-platform AI emotional support app with emotion-aware prompt design
Couple Chat Guard	React, TypeScript, Gemini API	Detects dismissive patterns in chat messages using AI sentiment analysis
Classical Apps MVP	React, Vite	Two interactive literary apps: immersive reading + branching narrative strategy game
Family Destiny App	Python	Traditional fortune calculation (八字/Bazi) with lunar calendar algorithms and chart visualization
Break the Loop	React, Vite	Micro-habit challenge app for breaking automated behaviors
Travel Journal	React, Vite	Personal travel logging and memory preservation app
Food Map App	HTML5, JavaScript	Personal restaurant discovery and review mapping application

EDUCATION

Taipei Medical University (TMU) — B.S. in Biomedical Engineering

LANGUAGES

- **Chinese (Mandarin):** Native proficiency
- **English:** Professional working proficiency (fluent written and verbal communication)

WORK PREFERENCES

- **Availability:** 40+ hours/week, flexible working hours (8:00 AM — 12:00 AM GMT+8)
- **Location:** Remote-first, open to global opportunities
- **Start Date:** Immediate

INTERESTS

Digital Health • AI Applications • EdTech • Interactive Storytelling • Cultural Heritage Technology • Product Innovation